

Raw scRNA-seq + TCR-seq

100,067 cells · 20,000+ genes · 4 patients

**Quality Control
Filtering**

Ambient RNA removal · doublet detection

Log-Normalization

Parallel Feature Engineering

PCA on 1,500 HVGs

50 principal components

TCR Sequence Encoding

k-mer + physicochemical features
≈ 74% cells with productive TCR

Integrated Feature Matrix

4 feature families · 29–429 features per set

